

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404789
Kind Code	B1
Date of Patent	September 02, 2025
Inventor(s)	Lyons; Timothy M.

Diesel exhaust fluid cooling system

Abstract

A vehicle includes a diesel engine having a crankcase and a crankcase ventilation system connected to the crankcase and configured to ventilate crankcase exhaust gases. The vehicle also includes a diesel exhaust fluid (DEF) reservoir including DEF therein, the DEF configured to reduce emissions of the diesel engine, and a DEF cooling system operably connected to the crankcase and the DEF reservoir. The DEF cooling system is configured to receive the DEF and use the DEF to cool the crankcase exhaust gases passing therethrough.

Inventors:	Lyons; Timothy M. (Batavia, IL)
Applicant:	International Engine Intellectual Property Company, LLC (Lisle, IL)
Family ID:	96812753
Assignee:	International Engine Intellectual Property Company, LLC (Lisle, IL)
Appl. No.:	18/587337
Filed:	February 26, 2024

Publication Classification

Int. Cl.: **F01M13/04** (20060101); **F01M13/00** (20060101); **F01M13/02** (20060101)

U.S. Cl.:

CPC **F01M13/04** (20130101); F01M2013/0038 (20130101); F01M2013/026 (20130101); F01M2013/0411 (20130101); F01N2610/02 (20130101)

Field of Classification Search

CPC: F01M (13/00); F01M (13/04); F01M (2013/0038); F01M (2013/026); F01M (2013/0411); F01N (2610/02); F01N (2610/10); B01D (2251/2062); B01D (2251/2067)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2057001	12/1935	Bogdany	123/573	F01M 13/04
2762349	12/1955	Malcolm	N/A	N/A
2771065	12/1955	Malcolm	N/A	N/A
2818052	12/1956	Trainer, Jr.	N/A	N/A
2902012	12/1958	Malcolm	N/A	N/A
4050237	12/1976	Pall	55/486	B01D 46/0031
4768493	12/1987	Ohtaka	123/573	F01M 13/00
5878731	12/1998	Bedkowski	123/573	F01M 13/04
6752856	12/2003	Graze, Jr.	55/DIG.19	B01D 50/20
7752840	12/2009	Stewart	N/A	N/A
8434434	12/2012	Adelman et al.	N/A	N/A
8485145	12/2012	Palumbo	N/A	N/A
10287943	12/2018	Schlitz	N/A	N/A
2005/0092309	12/2004	Bedkowski	123/572	F02M 25/06
2009/0199794	12/2008	Slaughter	123/41.86	F01M 13/04
2011/0083625	12/2010	Lyons	N/A	N/A
2015/0231564	12/2014	Wittrock	422/171	F01N 3/2066
2015/0231565	12/2014	Wittrock	423/212	F01N 13/009
2016/0025048	12/2015	Shiota	60/599	F02M 26/06
2017/0030247	12/2016	Suetou	N/A	F01N 3/08
2019/0242327	12/2018	Stanley	N/A	F02F 1/242
2020/0217230	12/2019	Hase	N/A	F02M 35/162
2021/0262372	12/2020	Hogan	N/A	B01D 53/9431
2022/0090548	12/2021	Wolk	N/A	F02D 41/3011
2023/0086242	12/2022	Scarf	60/605.1	F01M 13/028

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109715926	12/2018	CN	N/A
112021003669	12/2022	DE	N/A
2017141766	12/2016	JP	N/A
2020026763	12/2019	JP	N/A
WO-2019186968	12/2018	WO	N/A

Primary Examiner: Zaleskas; John M

Background/Summary

BACKGROUND

(1) This disclosure relates to a diesel engine system. Particularly, this disclosure relates to a system for removing particulates from crankcase exhaust gases to reduce emissions and improve engine performance.

RELATED ART

(2) Modern day legislative requirements to reduce emissions have effectively required diesel engine systems to include a closed crankcase ventilation system to process the crankcase exhaust gases. Crankcase exhaust gases, commonly referred to as “blow-by,” are the gases that are forced out of the crankcase during engine operation. Blow-by is a mixture of several components, including oil aerosol, soot particles, water vapor, hydrocarbons, and aldehydes. It was once common to include breathers equipped with road tubes to expel all of these components to the atmosphere. Nowadays, with closed crankcase ventilation systems, these components must be filtered out before recirculating the crankcase exhaust into the engine. Otherwise, any oil aerosols, or other particulates, can buildup on the housing and various other surfaces of the compressor stage of the turbocharger, which will eventually degrade the operation of the compressor and hinder engine performance.

(3) Crankcase ventilation systems perform the functions of filtering, collecting oil, and pressure regulation for regulating the pressure within the crankcase. Closed crankcase ventilation systems generally include a housing with an exhaust gas inlet and outlet, a pressure regulator, one or more filters disposed within the housing for capturing the various particulates suspended in the crankcase exhaust, and an outlet valve for the removal of the captured components. The filters perform the tasks of coalescing and separating the unwanted components, e.g., the oil mist particles, soot, and liquid volatiles created during the combustion process. The housing can serve as a sump chamber to collect liquid oil, which may be subsequently recirculated out through the outlet valve and back into the crankcase. However, despite improvements in crankcase ventilation systems, it can still be difficult to capture vapors and other particulates. Over time, these uncaptured particles will persist in collecting on the surfaces of the turbocharger compressor stage resulting in loss of compressor efficiency, damage, and reduced engine performance.

SUMMARY

(4) According to one embodiment, a vehicle includes a diesel engine having a crankcase and a crankcase ventilation system connected to the crankcase and configured to ventilate crankcase exhaust gases. The vehicle also includes a diesel exhaust fluid (DEF) reservoir including DEF therein, the DEF configured to reduce emissions of the diesel engine, and a DEF cooling system operably connected to the crankcase and the DEF reservoir. The DEF cooling system is configured to receive the DEF and use the DEF to cool the crankcase exhaust gases passing therethrough.

(5) According to another embodiment, a diesel engine system includes a diesel engine having a crankcase and a crankcase ventilation system connected to the crankcase and configured to ventilate crankcase exhaust gases. The diesel engine system further includes a diesel exhaust fluid (DEF) reservoir including DEF therein, the DEF configured to reduce emissions of the diesel engine, and a DEF cooling system operably connected to the crankcase and the DEF reservoir. The DEF cooling system configured to receive the DEF and use the DEF to cool the crankcase exhaust gases passing therethrough.

(6) According to yet another embodiment, a method for reducing emissions of a diesel engine includes several steps. The method includes an initial step of providing a diesel engine having a crankcase, a crankcase ventilation system connected to the crankcase, a diesel exhaust fluid (DEF) reservoir including DEF therein, the DEF configured to reduce emissions of the diesel engine, and a DEF cooling system operably connected to the crankcase and the DEF reservoir. The method further includes a step of ventilating, by the crankcase ventilation system, crankcase exhaust gases exiting from the crankcase. The method further includes a step of supplying the DEF to the DEF cooling system. The method further includes a step of cooling the crankcase exhaust gases, by the DEF cooling system, to condense vapors in the crankcase exhaust gases passing through the DEF cooling system so that the crankcase ventilation system filters particulate matter out of the crankcase exhaust gases to further reduce emissions of the diesel engine.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a graphical representation of a diesel exhaust fluid cooling system for cooling the crankcase gases or “blow-by” gases;
- (2) FIG. 2 is a graphical representation illustrating how the diesel exhaust fluid, acting as a working fluid, cools the crankcase exhaust gases;
- (3) FIG. 3 is a graphical representation of another embodiment of a diesel exhaust fluid cooling system, wherein the cooling system is positioned downstream of the crankcase ventilation system;
- (4) FIG. 4 is a graphical representation of another embodiment of a diesel exhaust fluid cooling system, wherein the cooling system is integrated into the crankcase ventilation system as a single unit; and
- (5) FIG. 5 is a graphical representation of another embodiment of a diesel exhaust fluid cooling system, wherein the cooling system transfers heat from the turbocharger compressor stage to the DEF.

DETAILED DESCRIPTION

- (6) Embodiments described herein relate to arrangements, and method of implementing, of a vehicle engine system for efficiently cooling crankcase exhaust gases with diesel exhaust fluid to further reduce emissions of the engine and prevent a buildup of particulate matter on engine components. The apparatus and method may be applied to various types of engine systems, including engine systems of generators, unmanned vehicles, submarines, or vehicles, including passenger vehicles, commercial vehicles, and recreational vehicles, such as highway or semi-tractors, straight trucks, buses, fire trucks, motorhomes, and etcetera. It is contemplated that the apparatus and method may be applied to vehicles having drivetrains including a diesel, gasoline, or gaseous fuel engine, as well as to vehicles having hybrid electric drivetrains. It is further contemplated that the apparatus and method may be applied to vehicles having manual transmissions, automatic transmissions, automated manual transmissions, continuously variable transmissions, hybrid electric transmissions, and hydraulic transmissions, as non-limiting examples. It is further contemplated that the apparatus and method may be applied to autonomous or semi-autonomous vehicles.
- (7) Turning now to FIGS. 1 and 2, a vehicle **10** implementing an embodiment of the present diesel engine system **12** is shown. The vehicle **10** can be in the form of a tow vehicle **10** for transporting a payload. For instance, the vehicle **10** can be a class **8** truck, capable of towing a trailer attached thereto, that is driven by the diesel engine system **12**. The diesel engine system **12** generally includes a diesel engine **14**, a turbocharger **16** with a turbine **18** and a compressor **20**, a crankcase **22**, an exhaust gas system **24**, e.g., an aftertreatment system **24**, which utilizes diesel exhaust fluid (DEF), stored in a DEF reservoir **26**, to treat and reduce emissions of the exhaust gases, a crankcase ventilation system **28**, and a DEF cooling system **30** for cooling crankcase exhaust gases to further reduce emissions and a buildup of particulate matter on engine components.
- (8) A common issue with closed crankcase ventilation systems is the buildup of particulate matter, such as oil and soot, in the engine components, such as within the ventilation system itself and/or on various surfaces in the turbocharger compressor stage. Over time, the accumulation of these deposits contributes to a loss of compressor efficiency and generally degrades engine performance. This issue is exacerbated by the difficulty of removing the various vapors suspended in the hot crankcase exhaust gases. To help assist in removing the vapors, such as oil aerosol particulates, the DEF cooling system **30** is designed to cause the condensation of the vapors, which thereby makes the condensed particulate matter far easier to remove. The particulate matter can be removed by the DEF cooling system **30** and/or by the crankcase ventilation system **28**. Thus, the DEF cooling system **30** is configured to cool the crankcase exhaust gases and condense oil aerosol particulates to

reduce emissions and a buildup of particulate matter on the compressor **20** to prevent compressor degradation caused by the buildup of particulate matter on the compressor **20**.

(9) The exhaust gas aftertreatment system **24** is configured to treat the exhaust of the diesel engine **14** for reducing the emissions thereof. The exhaust gas aftertreatment system **24** can include a catalytic reduction system including one or more diesel oxidation catalysts (DOCs), diesel particulate filters (DPFs), selective catalytic reduction (SCR) catalysts, a DEF injector for injecting the DEF, which can be ammonia or a urea solution, and various sensors for monitoring the exhaust gases.

(10) The crankcase ventilation system **28** is operably connected to the crankcase **22** and is configured to ventilate crankcase exhaust gases. The crankcase ventilation system **28** may generally include a housing with an inlet and an outlet, a pressure regulator, one or more filters, and a drain valve. In some embodiments, the crankcase ventilation system **28** can be a closed system. In some embodiments, the crankcase ventilation system **28** can be open and vent to the atmosphere.

(11) The DEF cooling system **30** is operably connected to the crankcase **22** and the DEF reservoir **26**. As shown, the DEF cooling system **30** is operationally connected in between the crankcase **22** and the crankcase ventilation system **28**. The DEF cooling system **30** is configured to receive the DEF and use the DEF to cool the crankcase exhaust gases passing therethrough. More particularly, the DEF cooling system **30** is configured for condensing various vapors in the crankcase exhaust gases, such as oil aerosol particulates, thus allowing the crankcase ventilation system **28** to more efficiently and easily filter particulate matter out of the crankcase exhaust gases.

(12) As shown in FIG. 2, the DEF cooling system **30** can include a housing **32** wherein the heat transfer from the crankcase gases to the DEF occurs. The housing **32** can include inlets and outlets **34, 36, 38, 40** for the DEF and crankcase exhaust gases, respectively, and an oil condensate outlet **42** which allows oil and other particulate matter to exit therethrough. In some embodiments, the collected oil may be returned to the crankcase **22** via a return line.

(13) In operation, hot crankcase gases, with oil aerosol therein, enter the DEF cooling system **30** at one end thereof. A DEF pump **44** will supply the DEF to the DEF cooling system **30**. As the crankcase gases pass through the DEF cooling system **30**, the cooler DEF receives the heat from the hot crankcase gases and accordingly cools the crankcase gases enough for causing the condensation of the oil aerosol. Thereafter, the now-condensed oil, which can be in liquid form, can exit through the outlet **42** and/or continue onto the crankcase ventilation system **28** for the capture and removal thereof. The heated DEF will return to the DEF reservoir **26**.

(14) In one embodiment, the DEF cooling system **30** can include a housing, an internal chamber, and one or more DEF fluid passageways disposed within the housing, thus allowing the hot crankcase exhaust gases to flow through the housing and around the DEF fluid passageways to accordingly cool the crankcase exhaust gases. In another embodiment, the DEF cooling system **30** can comprise a plurality of fins which additionally help to cool the crankcase exhaust gases as the gases pass therearound. In another embodiment, the DEF cooling system **30** can comprise one or more fluid passageways for the crankcase exhaust which are submerged in a reservoir of DEF within the housing of the DEF cooling system **30**.

(15) In one embodiment, the DEF pump **44** is configured as a dual purpose pump for dually supplying DEF to the exhaust gas treatment system **24** and the DEF cooling system **30**. Hence, the single pump **44** is fluidly connected to the DEF reservoir **26** and configured to supply the DEF to the exhaust gas system **24** and the DEF cooling system **30**. In another embodiment, two or more pumps can separately supply the DEF fluid to the exhaust gas treatment system **24** and the DEF cooling system **30**.

(16) Optionally, in some embodiments, the DEF cooling system **30** can further include a heat exchanger **46** in the return line for cooling the DEF before it returns to the DEF reservoir **26** by the pump (as shown in phantom in FIG. 1). The heat exchanger **46** can be in the form of any desired heat exchanger.

(17) Referring now to FIG. 3, there is shown another embodiment of a diesel engine system **112**. The diesel engine system **112** can be substantially similar to the diesel engine system **12** as discussed above except that the DEF cooling system **130** is operationally located at the compressor stage, after the crankcase ventilation system **28**. In other words, the DEF cooling system **130** is operationally connected in between the crankcase ventilation system **28** and the compressor **20**.

(18) Referring now to FIG. 4, there is shown another embodiment of a diesel engine system **212**. The diesel engine system **212** can be substantially similar to the diesel engine system **12** as discussed above except that the DEF cooling system **30** is integrated into the crankcase ventilation system **28** as one crankcase ventilation and cooling system **230**.

(19) Turning now to FIG. 5, there is shown another embodiment of a DEF cooling system **330**. The embodiment of the DEF cooling system shown in FIG. 5 may be used in combination with any of the embodiments of DEF cooling systems described previously herein, or may be used independently from any of the embodiments of the DEF cooling systems described previously herein. The embodiment of the DEF cooling system **330** shown in FIG. 5 is provided with a diffuser heat exchanger **350** that adjoins the compressor diffuser **352** of the compressor (not shown) of a turbocharger (not shown) of a diesel engine system. The diffuser heat exchanger **350** is provided with a DEF inlet **354** and a DEF outlet **356**, and acts as a heat exchanger between the compressor diffuser **352** and the DEF. In this way, the embodiment of the DEF cooling system **330** shown in FIG. 5 utilizes the DEF as a working fluid to transfer heat from the compressor of the turbocharger to the DEF storage tank (not shown).

(20) In so doing, the temperature of any oil mist particles, soot, or liquid volatiles will be kept below the coking temperature of the oil and oil additives. Moreover, since the DEF is at a lower temperature than the engine coolant, no secondary coolant loop is necessary to accomplish effective heat transfer. In addition, such vehicles already have a DEF pump that, with slight modifications, can be used to pump the working fluid of the present invention. Nevertheless, if needed, a heat exchanger may added in the DEF loop to control temperature of the DEF returning to the DEF tank.

(21) While illustrative arrangements, and control logic therefor, implementing the engine system have been described with respect to at least one embodiment, the arrangements and methods can be further modified within the spirit and scope of this disclosure, as demonstrated previously. This application is therefore intended to cover any variations, uses, or adaptations of the arrangement and method using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which the disclosure pertains and which fall within the limits of the appended claims.

Claims

1. A vehicle, comprising: a diesel engine having a crankcase; a crankcase ventilation system downstream of the crankcase and configured to ventilate crankcase exhaust gases; a diesel exhaust fluid (DEF) reservoir including DEF therein, the DEF configured to reduce emissions of the diesel engine; a DEF cooling system operably connected to the crankcase and the DEF reservoir, the DEF cooling system configured to receive the DEF and use the DEF to cool the crankcase exhaust gases passing there through; a turbocharger with a compressor connected to the diesel engine, the compressor receiving intake air and recirculated crankcase exhaust gases; and a DEF heat exchanger adjoining the compressor of the turbocharger, the DEF heat exchanger configured to receive the DEF and use the DEF to cool the compressor of the turbocharger.
2. The vehicle of claim 1, wherein: the DEF cooling system is configured to cool the crankcase exhaust gases and condense vapors therein, allowing the crankcase ventilation system to filter particulate matter out of the crankcase exhaust gases.
3. The vehicle of claim 1, wherein: the DEF heat exchanger further comprises a diffuser heat

exchanger that adjoins a diffuser of the compressor of the turbocharger.

4. The vehicle of claim 1, wherein: the DEF cooling system is configured to cool the crankcase exhaust gases and condense oil aerosol particulates to reduce emissions and a buildup of particulate matter on the compressor to prevent compressor degradation otherwise caused by the buildup of particulate matter on the compressor.
5. The vehicle of claim 4, wherein: the DEF cooling system is operationally connected in between the crankcase ventilation system and the compressor.
6. The vehicle of claim 1, wherein: the DEF cooling system is operationally connected in between the crankcase and the crankcase ventilation system.
7. The vehicle of claim 1, wherein: the DEF cooling system is integrated into the crankcase ventilation system.
8. The vehicle of claim 1, further comprising: a DEF pump fluidly connected to the DEF reservoir and configured to supply the DEF to an exhaust gas system and the DEF cooling system.
9. The vehicle of claim 8, wherein: the DEF cooling system further comprises a heat exchanger configured to cool the DEF before the DEF is returned to the DEF reservoir by the DEF pump.
10. A diesel engine system, comprising: a diesel engine having a crankcase; a crankcase ventilation system downstream of the crankcase and configured to ventilate crankcase exhaust gases; a diesel exhaust fluid (DEF) reservoir including DEF therein, the DEF configured to reduce emissions of the diesel engine; and a DEF cooling system operably connected to the crankcase and the DEF reservoir, the DEF cooling system configured to receive the DEF and use the DEF to cool the crankcase exhaust gases passing there through; a turbocharger with a compressor connected to the diesel engine, the compressor receiving intake air and recirculated crankcase exhaust gases; and a DEF heat exchanger adjoining the compressor of the turbocharger, the DEF heat exchanger configured to receive the DEF and use the DEF to cool the compressor of the turbocharger.
11. The diesel engine system of claim 10, wherein: the DEF cooling system is configured to cool the crankcase exhaust gases and condense vapors therein, allowing the crankcase ventilation system to filter particulate matter out of the crankcase exhaust gases.
12. The diesel engine system of claim 10, wherein: the DEF heat exchanger further comprises a diffuser heat exchanger that adjoins a diffuser of the compressor of the turbocharger.
13. The diesel engine system of claim 10, wherein: the DEF cooling system is configured to cool the crankcase exhaust gases and condense oil aerosol particulates to reduce emissions and a buildup of particulate matter on the compressor to prevent compressor degradation otherwise caused by the buildup of particulate matter on the compressor.
14. The diesel engine system of claim 13, wherein: the DEF cooling system is operationally connected in between the crankcase ventilation system and the compressor.
15. The diesel engine system of claim 10, wherein: the DEF cooling system is operationally connected in between the crankcase and the crankcase ventilation system.
16. The diesel engine system of claim 10, wherein: the DEF cooling system is integrated into the crankcase ventilation system.
17. The diesel engine system of claim 10, further comprising: a DEF pump fluidly connected to the DEF reservoir and configured to supply the DEF to an exhaust gas system and the DEF cooling system.
18. The diesel engine system of claim 17, wherein: the DEF cooling system further comprises a heat exchanger configured to cool the DEF before the DEF is returned to the DEF reservoir by the DEF pump.
19. A method, comprising: providing a diesel engine having a crankcase, a crankcase ventilation system downstream of the crankcase, a diesel exhaust fluid (DEF) reservoir including DEF therein, the DEF configured to reduce emissions of the diesel engine, a DEF cooling system operably connected to the crankcase and the DEF reservoir; a turbocharger with a compressor receiving intake air and recirculated crankcase exhaust gases and a DEF heat exchanger adjoining the

compressor configured to use the DEF to cool the compressor; ventilating, by the crankcase ventilation system, crankcase exhaust gases exiting from the crankcase; supplying the DEF to the DEF cooling system; and cooling the crankcase exhaust gases, by the DEF cooling system, to condense vapors in the crankcase exhaust gases passing through the DEF cooling system so that the crankcase ventilation system filters particulate matter out of the crankcase exhaust gases to further reduce emissions of the diesel engine.
